



2.6 Rational functions (HL)

Functions · IB Diploma · Concept guide and original exam-style practice

Understand the concept

A rational function of the form $f(x) = (ax+b)/(cx+d)$ ($c \neq 0$, $ad-bc \neq 0$) is a transformation of $y=1/x$. It has a vertical asymptote at $x=-d/c$ (where the denominator is zero) and a horizontal asymptote at $y=a/c$ (the ratio of leading coefficients, from polynomial division as $x \rightarrow \pm\infty$). The graph consists of two branches approaching these asymptotes.

Worked example: follow the reasoning

Find the asymptotes of $f(x) = (2x+1)/(x-3)$, and the intercepts.
Vertical asymptote: $x-3=0$, so $x=3$.
Horizontal asymptote: $y=2/1=2$ (ratio of leading coefficients).
Y-intercept: $f(0)=1/(-3)=-1/3$.
X-intercept: $2x+1=0$, $x=-1/2$.

Quick check · multiple choice

The function $f(x) = (3x-2)/(x+4)$ has vertical and horizontal asymptotes:

- A $x=4$, $y=3$
- B $x=-4$, $y=3$
- C $x=-4$, $y=-2$
- D $x=4$, $y=-2$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Consider $f(x) = (5-2x)/(x+1)$, $x \neq -1$. [7]
- (a) Write $f(x)$ in the form $A + B/(x+1)$ by division, and hence state the equations of both asymptotes.
 - (b) Find the coordinates of the point(s) where $f(x)$ crosses the coordinate axes.
 - (c) Sketch the key features described, and state the range of f .

Final answer: _____



Answers and explanation

Question 1 [7]

- (a) $(5-2x)/(x+1) = (-2(x+1)+7)/(x+1) = -2 + 7/(x+1)$. So asymptotes: vertical $x=-1$, horizontal $y=-2$.
- (b) Y-intercept: $f(0)=5/1=5$, point $(0,5)$. X-intercept: $5-2x=0$, $x=2.5$, point $(2.5,0)$.
- (c) Since $f(x)=-2+7/(x+1)$ and $7/(x+1)$ can take any nonzero real value, range is $f(x) \neq -2$ (all reals except -2).

Quick check answer

B. Vertical asymptote where denominator=0: $x=-4$. Horizontal asymptote: ratio of leading coefficients = $3/1 = 3$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.